

Could a rule you already know do this better?

No single deterministic rule reliably predicts weld defects from process parameters alone.

What would the answer let somebody do?

Let quality control prioritize which welds get inspected first or get 100% inspection versus sampling.

How would you know if you were wrong?

Compare the prediction against the actual X-ray / UT inspection result for that weld which is already being recorded as part of standard quality control.

What does a mistake cost here?

High - a missed defect in a structural weld can lead to catastrophic failure.

Problem 3: Time to failure of an HVAC compressor from vibration data.

This sorted as NUMBER.

What are you actually trying to find?

A continuous value - estimated remaining days/hours until failure predicted from vibration signature, temperature and run-hour sensor data.

Who measures that and how often?

Vibration and temperature sensors can log continuously which is dense enough data for this kind of model.

Could a rule you already know do this better?

No simple rule reliably predicts remaining life from raw vibration signatures. Pattern degradation over time is the exactly the kind of relationship ML is suited to versus a fixed maintenance interval rule which is often overly conservative or too late.

What would the answer let somebody do?

Let facilities staff schedule maintenance before failure instead of on a fixed calendar interval.



How would you know if you were wrong?

Compare predicted failure windows against actually actual logged failure or tear-down inspection dates as they occur over time.

What does a mistake cost here?

Moderate - a wrong prediction causes either an unplanned outage or wasted maintenance spend neither of which is safety-critical on its own.

Problem 4: Classify soil type from borehole log descriptions/photos.

This is sorted as CATEGORY:

What are you actually trying to find?

A discrete class - clay, silt, sand, gravel, fill or a USCS classification symbol predicted from field description text, photos or laboratory gradation data.

Who measures that and how often?

Geotechnical field engineers log this at every borehole / sample interval already per standard field logging procedure.

Would a rule you already know do this better?

There are existing classification rules but applying them consistently from photos or inconsistent field notes is where human judgement varies significantly, a model can help standardize that judgement call, not replace the classification system itself.

What would the answer let somebody DO?

Let a geotechnical report get drafted faster and more consistently and flag borings where the classification is uncertain for a senior engineer to review.

How would you know if you were wrong?

Compare the predicted classification against lab-confirmed gradation for the same sample.



What does a mistake cost here?

Low to moderate on its own - misclassification would typically get caught by downstream lab testing before it reaches a design decision.

Problem 5: Determine the legal load rating of an existing bridge.

This is sorted as NOT A MACHINE LEARNING PROBLEM.

What are you actually trying to find?

A specific code-defined value but one governed entirely by structural analysis and AASHTO Manual for Bridge Evaluation procedures.

Who measures that and how often?

Bridge engineers calculate this through structural analysis during periodic inspections.

Would a rule you already know do this better?

Yes decisively - the load rating is deterministic calculation with legal standing. A statistically predicted rating would carry no code compliance or liability protection so the existing rule is strictly better here.

What would the answer let somebody DO?

Let a state post correct weight limits and permit oversize loads.

How would you know if you were wrong?

A calculation or input error checked through peer review and code compliance.

What ~~mista~~ does a ~~mista~~ke cost here?

Very high - an incorrect rating can permit an overweight load onto a bridge that can't safely carry it, risking structural failure and loss of life.